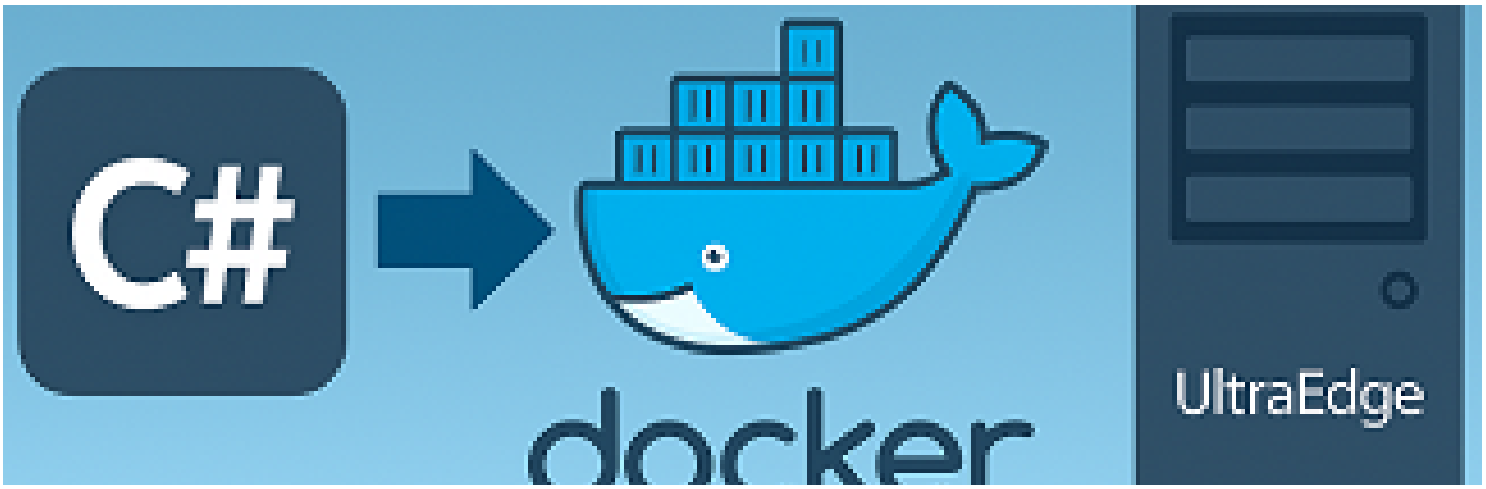


How to send a Docker file to the UltraEdge from a C# Application

Reference : RA 449

Description

This Reference Architecture describes how to send a Docker file to the UltraEdge from a C# application.

While this operation is usually demonstrated from an IGXL test program, it can also be performed from a separate standalone application.
Any language can be used to implement such an application. We show here how to do this using C#.

Several code samples are provided according to the following scenarios:

- Encrypted or non-encrypted docker files
- .Net Framework 4.8 or .Net 8

General Technical Info

- **Language:** C# - .Net Framework 4.8 and .Net 8
- **Environment:** Windows

General sequence

The following sequence can be used to deploy a non-encrypted docker file to the UltraEdge:

- Connect
- Start a session with no session tag (the parameter is an empty string)
- Send the docker file to the UltraEdge
- Load the docker image
- Execute the container

To manage an encrypted docker file, the sequence is slightly modified:

- Connect
- Start a session with no session tag (the parameter is an empty string)
- Send the encrypted docker file and additional key file to the UltraEdge
- Decrypt the docker file
- Load the docker image
- Execute the container

Function to load and run a docker

The following code sample provides a method to copy a docker file to the UltraEdge, load the docker image and run the container.
Here is a description of the parameters:

Parameter name	Description
`dockerfilepath`	Full path of the docker file to send to the UltraEdge
`imagename`	Docker image name used in the code. It may be different from the actual image name loaded in the UltraEdge.
`appname`	Application name used in the code. Any name can be chosen.
`portMapping`	Mapping from docker network port to the UltraEdge network port. These are valid for applications working inbound (for example a socket server) and outbound (a client).
`dockerCommandLine`	Main application to run in the docker container.
`workingDir`	Parent folder of an `exec` subfolder automatically mounted to exchange files between the docker and the UltraEdge RAM disk. May be an empty string (in which case, the `exec` subfolder will be created in `/app` on the docker filesystem)
`session_tag`	Used for encryption. This is the tag name specified when the private key was installed.

```
public void LoadDockerFile(
    string dockerfilepath, string imagename, string appname,
    string portMapping, string dockerCommandLine, string workingDir, string session_tag
= "")
{
    string dockerfilename = new FileInfo(dockerfilepath).Name;
```

```
    // Connect to the UltraEdge
    UE.Connect();
    printError("Connect");

    // Start a session
    UE.SecureSession.Start(session_tag);
    printError("Start session");

    if (String.IsNullOrEmpty(session_tag))
    {
        // Send Docker file
        UE.WriteFile(dockerfilepath, dockerfilename);
        printError("Write docker file");
    }
    else
    {
        // Send Docker encrypted files
        UE.WriteFile(dockerfilepath + ".enc", dockerfilename + ".enc");
        printError("Write docker encrypted file");
        UE.WriteFile(dockerfilepath + ".key.enc", dockerfilename + ".key.enc");
        printError("Write encryption key file");

        // Decrypt the docker file
        UE.DecryptFile(dockerfilename + ".enc");
        printError("Docker file decryption");
    }

    // Load docker image
    UE.LoadDockerImage(imagename, dockerfilename);
    printError("Load docker image");

    // Run docker container
    UE.Application(appname).RunDockerContainer(workingDir, dockerCommandLine,
imagename, portMapping);
    printError("Run docker container");
}
```

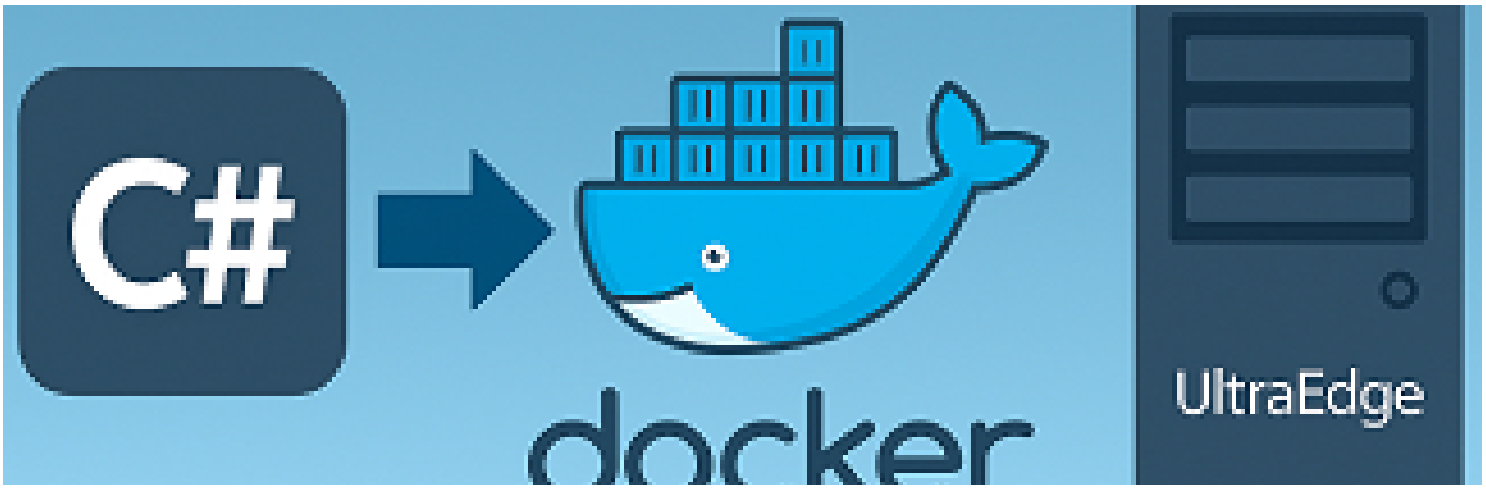
.Net environments

In this reference architecture, two sample projects are provided: .Net Framework 4.8 and .Net 8. As can be observed in the sample source code provided, the same .cs files are used for both environments. There are however slight differences between .Net Framework 4.8 and .Net 8.

Here are the important points to setup a C# project to work with the UltraEdge library.

1. The C# project must target x64

2. A reference to `UltraEdge.dll` must be added
3. The `UEProxy.cs` takes care of resolving the DLL references.
4. **.Net 8 only**: the NuGet package `System.Configuration.ConfigurationManager` must be installed. This package was integrated to .Net Framework but has been removed by .Net.
5. **.Net 8 only**: when calling `Disconnect()`, make sure to catch the `PlatformNotSupportedException` exception to avoid crashing the application. This is due to the use of `Thread.Abort()` which only supported on .Net Framework applications.



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More details about those source code here

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